

Technical Assessment: End-to-End Assessment Automation

Candidate Interview Assignment | Engineering & Automation Pipeline

1. Overview

We are building a scalable, automated pipeline to streamline the creation, parsing, and publishing of curriculum assessments. Our curriculum architecture spans N courses, each consisting of M units, X chapters, and itemized question banks.

Your objective is to design and build an end-to-end automation tool that leverages LLM APIs, document processing utilities, and custom integration scripts to convert learning planner outcomes into live portal assessments with minimal manual intervention.

2. Core Requirements & Workflow

Phase 1: Intelligent Question Generation & Export

- **Inputs:** Learning outcomes, planner documentation, and target grade levels.
- **Generation Engine:** Automatically generate high-quality question banks controlled by:
 - **Difficulty Scaling:** Balanced ratio of Easy, Medium, and Hard items.
 - **Cognitive Taxonomy:** Testing diverse skills (e.g., Recall, Application, Critical Thinking based on Bloom's Taxonomy).
 - **Age Appropriateness:** Tone, vocabulary, and context adapted specifically to the target grade level.
- **Document Compilation:** Auto-format and compile the generated questions into a structured document format (e.g., PDF/DOCX, including image placeholders or rendered visual aids).

Phase 2: Ingestion Pipeline (Doc-to-Portal Automation)

- **Automated Parsing:** Programmatically extract text, structured schema, and embedded media/images from the generated document.
- **Portal Publishing:** Implement direct API integration to automatically ingest parsed content into the assessment platform, completely eliminating manual copy-pasting.

3. Advanced Logic & System Design (Stretch Goals)

- **Teacher Dashboard Synchronization:** Auto-link published assessments to the teacher analytics panel upon release.
- **Immutability & Version Control:** Allow edits to live assessments while *hard-saving (locking)* historical score data for students who have already submitted attempts.
- **Dynamic Outcome Propagation:** Trigger pipeline alerts/updates to question banks whenever upstream learning outcomes are modified in the planner.

4. Submission Deliverables

Candidates must submit **one** of the following options:

- **Option A (Functional Prototype):** A working **Docker container** (packaged with clean source code) containing the automation tool/pipeline scripts that execute the end-to-end workflow.
- **Option B (System Blueprint Worksheet):** A comprehensive **PDF worksheet** detailing system architecture diagrams, API schemas, OCR/doc-parsing pipeline flow, prompt templates, and error-handling strategies.